

50E HDR Exposure/Gain Control

Presentation Subtitle

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HDR HTS

- Below is HDR mode HTS calculation.

Format	HTS
DCG	{0x380c, 0x380d}
DCG + VS	{0x380c, 0x380d} + {0x384c, 0x384d}
STG2	{0x380c, 0x380d} * 2
STG3	{0x380c, 0x380d} * 3

- Exposure register unit is HTS. One HTS time is $HTS / 75MHz$.
- Exposure and VTS step is 2 for binning, 4 for full size.
- Full size minimum exposure is 8, binning minimum exposure is 4. Full size linear mode max exposure is VTS - 32. Binning linear format including DCG maximum exposure is VTS - 16.

SCG STG2

- Exp_L {0x3500, 0x3501, 0x3502}, Exp_M {0x3540, 0x3541, 0x3542}
- Gain_L {0x3508, 0x3509}, Gain_M {0x3548, 0x3549}
- Normal mode gain range 1 ~ 63.75x; High gain mode gain range 1 ~ 255x
- Exposure Requirement, the following (n+2), (n+1) is frame number that has effect.

Host sets exposure Exp_L(n+2), Exp_M(n+2), VTS(n+1) in frame n. Frame n+2 image data is for Exp_L(n+2), Exp_M(n+2). Exp_L(n+2) starts in frame n+1, Exp_M(n+2) starts in frame n+2.

Some platform has minimum gap requirement from L frame SOF to M frame SOF. So we can adjust register 0x3830 value to meet requirement. But it affects exposure formula. Below formula adds 0x3830 register value.

- **$\text{Exp_L}(n+2) + \text{Exp_M}(n+1) < \text{VTS}(n+1) - 32 - (0x3830 \text{ value} + 1) * 2$**

Non-overlap adds: $\text{Exp_M}(n+1) < \text{VTS}(n+1) - (\text{Image Height}) - 56 - (0x3830 \text{ value} + 1) * 2$

- Below condition makes sure M frame in frame n+1 doesn't conflict to M frame in frame n+2, overlap case
 $\text{Exp_M}(n+2) \geq \text{Exp_M}(n+1) + \text{Image Height} - \text{VTS}(n+1)$, otherwise increase VTS(n+1) to meet requirement.

SCG STG3

- Exp_L {0x3500, 0x3501, 0x3502}, Exp_M {0x3540, 0x3541, 0x3542}, Exp_S {0x3580, 0x3581, 0x3582}
- Gain_L {0x3508, 0x3509}, Gain_M {0x3548, 0x3549}, Gain_S {0x3588, 0x3589}
- Normal mode gain range 1 ~ 63.75x; High gain mode gain range 1 ~ 255x
- Exposure Requirement, the following (n+2), (n+1) is frame number that has effect.

Host sets exposure Exp_L(n+2), Exp_M(n+2), Exp_S(n+2), VTS(n+1) in frame n. Frame n+2 image data is for Exp_L(n+2), Exp_M(n+2), Exp_S(n+2). Exp_L(n+2) starts in frame n+1, Exp_M(n+2) and Exp_S(n+2) start in frame n+2.

Some platform has minimum gap requirement from L frame SOF to M frame SOF, from M frame SOF to S frame SOF. So we can adjust register 0x3830 value to meet requirement. But it affects exposure formula.

Below formula adds 0x3830 register value.

- **$\text{Exp_L}(n+2) + \text{Exp_M}(n+1) + \text{Exp_S}(n+1) < \text{VTS}(n+1) - 68 - (0x3830 \text{ value} + 1) * 4$**
Non-overlap adds: $\text{Exp_M}(n+1) < \text{VTS}(n+1) - (\text{Image Height}) - 116 - (0x3830 \text{ value} + 1) * 4$
- Below condition makes sure S frame in frame n+1 doesn't conflict to S frame in frame n+2, overlap case
 $\text{Exp_M}(n+2) + \text{Exp_S}(n+2) \geq \text{Exp_M}(n+1) + \text{Exp_S}(n+1) + \text{Image Height} - \text{VTS}(n+1)$, otherwise increase VTS(n+1) to meet requirement.

Full Crop STG2

- Exp_L {0x3500, 0x3501, 0x3502}, Exp_M {0x3540, 0x3541, 0x3542}
- Gain_L {0x3508, 0x3509}, Gain_M {0x3548, 0x3549}

• Normal mode gain range 1 ~ 15.9375x; High gain mode gain range 1 ~ 63.75x

- Exposure Requirement, the following (n+2), (n+1) is frame number that has effect.

Host sets exposure Exp_L(n+2), Exp_M(n+2), VTS(n+1) in frame n. Frame n+2 image data is for Exp_L(n+2), Exp_M(n+2). Exp_L(n+2) starts in frame n+1, Exp_M(n+2) starts in frame n+2.

Some platform has minimum gap requirement from L frame SOF to M frame SOF. So we can adjust register 0x3830 value to meet requirement. But it affects exposure formula. Below formula adds 0x3830 register value.

- **$\text{Exp_L}(n+2) + \text{Exp_M}(n+1) < \text{VTS}(n+1) - 49 - 0x3830 \text{ value}$**

Non-overlap adds: $\text{Exp_M}(n+1) < \text{VTS}(n+1) - (\text{Image Height}) - 105 - 0x3830 \text{ value}$

- Below condition makes sure M frame in frame n+1 doesn't conflict to M frame in frame n+2, overlap case

$\text{Exp_M}(n+2) \geq \text{Exp_M}(n+1) + \text{Image Height} - \text{VTS}(n+1)$, otherwise increase VTS(n+1) to meet requirement.

50E binning STG2/STG3, full crop STG2 setting change if need medium or short exposure time long.

- Beside STG2/STG3 VTS/exposure condition when STG2/STG3 medium exposure time is longer than image height, we need to set below register to make it working.

When STG3 short exposure time longer than (8100 - image height - medium exposure), it also needs below setting. For example 4096x3072 STG3 L_Exp = 2600, M_Exp = 2600, S_Exp > 2428.

6c 3400 08

6c 4010 28

6c 4506 01

6c 450c 00

DCG, DCG Combine

- Exp_DCG {0x3500, 0x3501, 0x3502}. For DCG combine set {0x3500, 0x3501, 0x3502} and {0x3540, 0x3541, 0x3542} to same value.
- Gain_HCG {0x3508, 0x3509}, Gain_LCG {0x3548, 0x3549}
- Normal mode gain range 1 ~ 15.9375x; High gain mode gain range 1 ~ 63.75x
- For DCG combine when change HCG/LCG gain, please use below setting. If don't use combine, then no need to set 0x5006, {0x5019[1:0], 0x501a, 0x501b} and {0x501c[1:0], 0x501d, 0x501e}.

6c 5006 06 ;format setting already includes this register, it is 0 without combine.

{0x3508, 0x3509} = HCG analog gain * 256 ;The gain is 8-bit integer and 8-bit fraction.

{0x5019[1:0], 0x501a, 0x501b} = HCG analog gain * 1024 * HCG digital gain ; The gain is 8-bit integer and 10-bit fraction.

{0x3548, 0x3549} = LCG analog gain * 256 ;The gain is 8-bit integer and 8-bit fraction.

{0x501c[1:0], 0x501d, 0x501e} = LCG analog gain * 1024 * LCG digital gain ; The gain is 8-bit integer and 10-bit fraction.

DCG+VS, DCGCombine+VS

- Exp_DCG {0x3500, 0x3501, 0x3502}, Exp_VS {0x3580, 0x3581, 0x3582}. For DCGCombine+VS set {0x3500, 0x3501, 0x3502} and {0x3540, 0x3541, 0x3542} to same value.
- Gain_HCG {0x3508, 0x3509}, Gain_LCG {0x3548, 0x3549}, Gain_VS {0x3588, 0x3589}
- Gain range 1 ~ 63.75x
- Exposure Requirement, the following (n+2), (n+1) is frame number that has effect.

Host sets exposure Exp_DCG(n+2), Exp_VS(n+2), VTS(n+1) in frame n. Frame n+2 image data is for Exp_DCG(n+2), Exp_VS(n+2). Exp_DCG(n+2) starts in frame n+1, Exp_VS(n+2) starts in frame n+2.

Some platform has minimum gap requirement from DCG frame SOF to VS frame SOF. So we can adjust register 0x3830 value to meet requirement. But it affects exposure formula. Below formula adds 0x3830 register value.

- $\text{Exp_DCG}(n+2) + \text{Exp_VS}(n+1) < \text{VTS}(n+1) - 32 - (0x3830 \text{ value} + 1) * 2$
Non-overlap or Combine adds: $\text{Exp_VS}(n+1) < \text{VTS}(n+1) - (\text{Image Height}) - 56 - (0x3830 \text{ value} + 1) * 2$
- Below condition makes sure VS frame in frame n+1 doesn't conflict to VS frame in frame n+2, no combine overlap case
No Combine: $\text{Exp_VS}(n+2) \geq \text{Exp_VS}(n+1) + \text{Image Height} - \text{VTS}(n+1)$, otherwise increase $\text{VTS}(n+1)$ to meet requirement.

DCG+VS, DCGCombine+VS

- For DCGCombine+VS when change HCG/LCG gain, please use below setting. If don't use combine, then no need to set 0x5006, {0x5019[1:0], 0x501a, 0x501b}, {0x501c[1:0], 0x501d, 0x501e}, and {0x501f[1:0], 0x5020, 0x5021}

6c 5006 06 ;format setting already includes this register, it is 0 without combine.

{0x3508, 0x3509} = HCG analog gain * 256 ;The gain is 8-bit integer and 8-bit fraction.

{0x5019[1:0], 0x501a, 0x501b} = HCG analog gain * 1024 * HCG digital gain ;The gain is 8-bit integer and 10-bit fraction.

{0x3548, 0x3549} = LCG analog gain * 256 ;The gain is 8-bit integer and 8-bit fraction.

{0x501c[1:0], 0x501d, 0x501e} = LCG analog gain * 1024 * LCG digital gain ;The gain is 8-bit integer and 10-bit fraction.

{0x3588, 0x3589} = VS analog gain * 256 ;The gain is 8-bit integer and 8-bit fraction.

{0x501f[1:0], 0x5020, 0x5021} = VS analog gain * 1024 * VS digital gain ;The gain is 8-bit integer and 10-bit fraction.

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